

Uttam Mahata

+91-9679244057 | mahatauttam0408@gmail.com | Medinipur, West Bengal, India
[LinkedIn](#) | github.com/Uttam-Mahata | uttamm.web.app | [Hashnode](#)

EDUCATION

Indian Institute of Engineering Science & Technology, Shibpur

B.Tech – Computer Science and Technology, **CGPA: 9.07 / 10.0** (Till 7th Semester)

Howrah, India

Nov. 2022 – May 2026

PROFESSIONAL EXPERIENCE

OpenLM India

Software Engineering Intern

Kolkata, India

Jan. 2026 – Present

- Optimized **Spark** workloads on **Kubernetes** by tuning CRD memory allocations and rightsizing resource requests across 30+ microservices, eliminating OOMKills; authored **4 Azure DevOps CI/CD pipelines**.
- Built an autonomous incident-response system using **Google ADK** and **MCP** that inspects logs, analyzes root causes via **Datadog**, and auto-creates **Azure DevOps** tickets – an intelligent SRE agent powered by **LLM**.
- Diagnosed and resolved container path ownership gaps in Dockerfile causing PermissionDenied failures in non-root environments.

Alpha AI Service Pvt Ltd

Software Development Intern

New Delhi, India (Remote)

Jun. 2025 – Aug. 2025

- Implemented **Factory Provider** design pattern for multi-provider AI support (**Gemini API**, **LangChain**), enabling image, TTS, and video generation across 10+ content types; built **Razorpay**-integrated usage-based billing system.
- Deployed highly-available **K3s** cluster on 4 VPS nodes with **WireGuard** full-mesh VPN, AES-CBC secret encryption, and external MySQL datastore.
- Provisioned CI/CD with 8 self-hosted GitHub Actions runners; authored **Ansible**-based **GitOps** engine deploying 23 microservices with **Infisical** secret sync and zero-touch rolling updates.

PROJECTS

DilithiAuth – Post-Quantum OAuth 2.0 Identity Provider

Mar. 2026 – Present

- Architected a post-quantum OAuth 2.0 / OpenID Connect (OIDC) identity provider that dual-signs every token with **Ed25519 + ML-DSA-44 (NIST FIPS 204)**, protecting against Harvest-Now-Decrypt-Later attacks where encrypted data is collected today and decrypted using future quantum computers.
- Engineered three token formats to manage large post-quantum signature size: compact Thin JWT (approx. 500B) with server-side storage, Jumbo JWT (3–4 KB), and binary **CBOR Web Token** (1.2 KB) – each selectable based on infrastructure constraints; implemented **Edge Proxy** gateway for centralized PQC verification.

RootAccess – CTF Competition Platform

Mar. 2026 – Present

- Architected and shipped ctf.rootaccess.live, a full-stack Capture The Flag platform using **Go (Gin)** API and **Angular 21** frontend with real-time scoreboards via **WebSocket** and **Upstash Redis** pub/sub; deployed multi-cloud infrastructure across **Google Cloud Functions**, **AWS Lambda**, and **Cloudflare Workers**.
- Implemented dynamic scoring (CTFd formula), **JWT** auth with HTTP-only cookies, per-IP rate limiting, email verification, and **role-based access control (RBAC)** for multi-team competition, delivering 3M free requests/month.

CourseWagon – Generative AI Course Authoring & LMS

Aug. 2024 – Present

- Shipped an enterprise-grade LMS that auto-generates full course hierarchies (subjects, chapters, and topics) using **Gemini API**, cutting manual curriculum-authoring effort by **60%**.
- Integrated **Firebase Auth + JWT**, **Oracle Cloud Object Storage** for media, KaTeX/MathJax for equations, and Mermaid.js for diagrams; deployed on **GCP** with **Google Cloud Run**.

SKILLS

Programming Languages: C, Java, Python, TypeScript, Go (Golang)

Frontend & Backend: Angular, React, Spring Boot, FastAPI, Gin (Go)

Databases: MySQL, PostgreSQL, MongoDB, Firebase Firestore, Redis, SQLite

Cloud & DevOps: Linux, Docker, Kubernetes, GCP, AWS, Azure DevOps, GitHub Actions, CI/CD

ACHIEVEMENTS

Brain Dead Hackathon – IEST Shibpur

2nd Place – LLM-based hybrid summarization model for research papers

Howrah, India

Mar. 2025

IEST-UCO Bank Hackathon

Built Aegis, a centralized framework for a secure financial environment

Howrah, India

Jul. 2025 – Aug. 2025